KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 2nd Year 1st Term Examination, 2022
ME 2115
(Fluid Mechanics)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a). What is the concept of continuum in defining fluid properties? 07

1(b). What is meant by viscosity? Explain the effect of temperature on the viscosity of liquid and gas. 10

1(c). Write short note on: 08

(i) Cavitation, (ii) Surface tension.

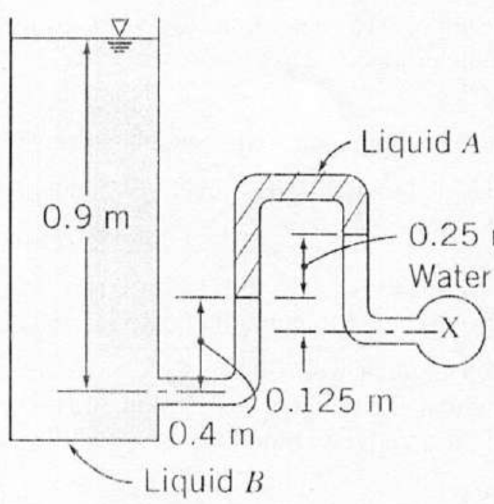
1(d). A 1.5 mm-diameter tube is inserted into an unknown liquid whose density is 960 kg/m³, and it is observed that the liquid rises 6 mm in the tube, making a contact angle of 15°. Determine the surface tension of the liquid. 10

2(a). Prove that the pressure is the same in all directions at a point in a static fluid. 09

2(b). State and explain the Pascal's law. 06

2(c). What is the difference between a barometer and a manometer? 05

2(d). Determine the gage pressure in kPa at a point X, if liquid A has SG = 1.3 and liquid B has SG = 0.8. The liquid surrounding point X ia water, and the tank on the left is open to the atmosphere. 15



3(a). What is meant by center of pressure? Derive an expression for the center of pressure on a submerged vertical plane surface. 13

3(b). What is meant by buoyant force? Derive an expression for the buoyant force acting on a body completely submerged in liquid. 08

3(c). Define the followings: 08

(i) Uniform flow, (ii) Compressible flow, (iii) Viscuss flow, and (iv) Path-line.

3(d) What is vorticity? Show the relation between circulation and vorticity. 06

4(a) What is Reynolds transport theorem? Derive Reynolds transport theorem and show the relation between system and control volume. 20

4(b) What is laws of conservation of mass? Derive continuity equation from the laws of conservation of mass. 15

SECTION – B

5(a). Derive the Bernoulli's equation from Euler's equation of motion. What are the limitations of this equation. 17

5(b) What are the major losses and minor losses in pipe? 08

5(c) What is stagnation pressure? Draw hydraulic grade line and energy grade line for two reservoirs connecting with an inclined pipe. 10

6(a) What is a venturimeter? Derive an expression for the discharge through a venturimeter. 15

6(b) Define the following: 09

(i) Vena-contracta, (ii) Co-efficient of contraction, (iii) Co-efficient of discharge.

6(c) A 30 cm diameter pipe conveying water, branches into two pipes of diameters 20 cm and 15 cm, respectively. If the average velocity in the 30 cm diameter pipe is 2.5 m/s, find the discharge in this pipe. Also determine the velocity in 15 cm pipe if the average velocity in 20 cm diameter pipe is 2 m/s. 11

7(a) What is nappe and sill? Distinguish between notches and weirs. 08

7(b) Define laminar sub-layer. Draw different level of boundary layers when a viscous fluid is flowing over a flat plate. 10

7(c) State the followings: 05

(i) Displacement thickness, (ii) Momentum thickness.

7(d) Water flows over a rectangular weir of 1 m wide at a depth of 150 mm and afterwards passes through a triangular right-angled weir. Taking C_d for the rectangular and triangular weir as 0.62 and 0.59, respectively. Find the depth over the triangular weir. 12

8(a) What is the difference between skin friction and pressure drag? Deduce an expression of Von-Karman momentum integral equation with zero pressure gradient. 20

8(b) A thin plate is moving in still atmospheric air at a velocity of 5 m/s. The length of the plate is 0.6 m and width of 0.5 m. Calculate (i) the thickness of the boundary layer at the end of the plate, and (ii) drag force on one side of the pipe. Take density of air as 1.24 kg/m^3 and kinematic viscosity 0.15 stokes. 15

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 2nd Year 1st Term Examination, 2022
EE 2113
(Electrical Machines)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a). Briefly explain the working principle of a dc generator. 10

1(b). Define pole pitch, front pitch and back pitch. Compare between lap and wave winding. 08

1(c). Explain different losses in dc generator. A 4 pole dc generator is delivering 20 A to a load of 10 Ω. The armature resistance is 0.5 Ω and the shunt field resistance is 50 Ω. The iron loss and friction loss are 1000 W and 500 W, respectively. Calculate the e.m.f and efficiency of the machine with allowing a drop of 1V per brush. 12

1(d). Show various power stages with their efficiency in case of a dc generator. 05

2(a). Why the parallel operation of shunt generators is necessary? Mention the conditions for paralleling dc generators. 10

2(b). Briefly explain important characteristics curves of a dc generator with neat sketch. Mention some applications of dc generator. 12

2(c). What is back e.m.f? Write down the voltage equation of dc motor and show the significance of back e.m.f. 07

2(d). Derive the condition for maximum efficiency of a dc motor, $E_b = \frac{V}{2}$, where symbols have their usual meaning. 06

3(a). "The core loss is practically the same under all load conditions" Justify the statement with proper illustration. 09

3(b). A 25 kW, 250 V dc shunt generator has armature and field resistances of 0.06 Ω and 100Ω, respectively. Determine the total armature power developed when working i) as a generator delivering 25 kW output, and ii) as a motor taking 25 kW input. 09

3(c). Why starter is necessary? What are the limitations of three point starter? How to solve them using four point starter? 10

3(d). What are the factors controlling motor speed? Briefly explain flux control method. 07

4(a). Show that the induced e.m.f per turn is the same in both the primary and seceondary windings of a transformer. Classify transformer based on cooling. 10

4(b). What are the parameters of a transformer? How to find them using open circuit test and short circuit test? Explain 11

4(c). Why transformer rating is given in kVA? Draw the exact and approximate equivalent circuit diagram of a transformer. 09

4(d). In a 25 kVA, 2000/200 V, single phase transformer, the iron and full load copper losses are 350 W and 400 W, respectively. Calculate the efficiency at unity power factor on full load. 05

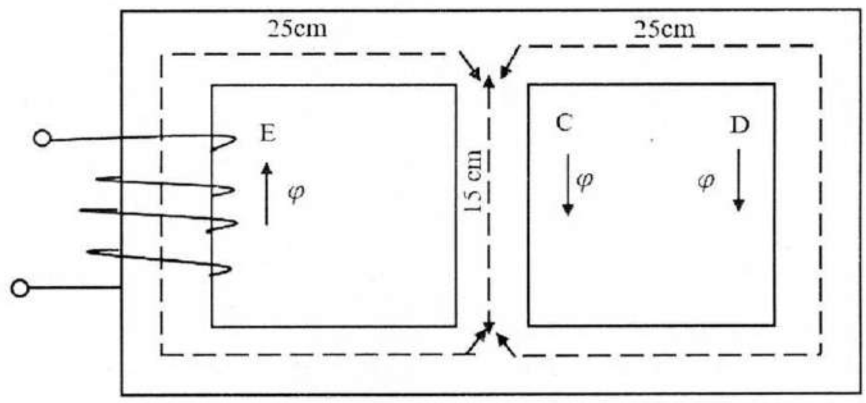
SECTION – B

5(a). State and explain Ampere's circuital law and find mmf along a long straight conductor. 07

5(b). Define flux density, permeability and reluctance. Compare between electrical circuit and magnetic circuit. 08

5(c). What is hysteresis loop? Draw a hysteresis loop with coercive force, residual magnetism and retentivity properly identified. 10

5(d). A cast steel magnetic structure for a bar of section $8cm \times 2cm$ is shown in figure. Determine the current that the 500 turn-magnetizing coil on the left limb should carry so that a flux of 2 mWb is produced in the right limb. Total $\mu_r = 600$ and neglect leakage. 10



6(a). How armature reaction affects the terminal voltage of alternator? Is there any dependency of armature reaction on the types of load connected to the alternators? Justify it. 10

6(b). Why the induction motor is called rotating transformer? 07

6(c). Find the equation of induced emf for an alternator. 07

6(d). A 60 kVA, 220V, 50Hz, single phase alternator has effective armature resistance of 0.016Ω and an armature leakage reactance of 0.07 Ω. Compute the voltage induced in the armature when the alternator is delivering rated current at a load power factor of i) Unity, ii) 0.7 lagging, and iii) 0.7 leading. 11

7(a). What is voltage regulation? Explain the synchronous impedance method of determining voltage regulation. 10

7(b). Why synchronous generators are operated in parallel? Mention the conditions required for parallel operation of synchronous generaors. 08

7(c). How rotating magnetic field is created in induction motor for three-phase supply? Show that resultant flux is 1.5 times the maximum value of the flux for three-phase supply. 10

7(d). Can an induction motor run at synchronous speed? Explain briefly. Mention some applications of induction motor. 07

8(a). Draw the approximate equivalent circuit of an induction motor. Average torque produced for an induction motor is maximum for resistive load. Explain with mathematical and graphical representation. 12

8(b). Find maximum torque under running condition for an induction motor. 12

8(c). Explain the torque speed characteristics of an induction motor. 05

8(d). A 150 kW, 3000 V, 50Hz, 6 pole star connected induction motor has a star connected slip ring rotor with a transformation ration of 3:6 (Stator/ rotor). The rotor resistance is 0.1Ω per phase and it's per phase leakage reactance is 3.61mH. The star impedance may be neglected. Find the starting current and starting torque on rated voltage with sort circuited slip rings. 06

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 2nd Year 1st Term Examination, 2022
Math 2113
(Linear Algebra and Vector Analysis)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a). What values of a and b, if any, will make the matrices $A = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & a \\ 4 & b \end{bmatrix}$ 08

(i) Commute, (ii) anti-commute?

1(b). Define Non-singular matrix. Find the inverse of the matrix $A = \begin{pmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{pmatrix}$, if exists. 12

1(c). Determine the rank of the matrix $A = \begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 \\ 3 & 4 & 3 & 5 & 5 & 7 \\ 3 & 6 & 4 & 9 & 10 & 11 \\ 1 & 2 & 4 & 3 & 6 & 9 \end{pmatrix}$ by reducing it to row canonical from and hence obtain its normal form. 15

2(a). Find the real numbers x, y, z such that $\begin{pmatrix} 3 & x+2i & yi \\ 3-2i & 0 & 1+zi \\ yi & 1-xi & -1 \end{pmatrix}$ is Hermitian 08

2(b). For what value(s) of 'a' does the following system of equations: 13

$$\left.\begin{aligned} x+2y+2z &= 1 \\ x+ay+3z &= 3 \\ x+4y+az &= 0 \end{aligned}\right\}$$ Provides (i) unique solution, (ii) no solution, (iii) infinitely many solutions?

2(c). Find the eigen values and corresponding eigen vectors of the following matrix, 14

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

Also, find the modal and spectral matrices of **A**, if exist.

3(a). Show that the set of vectors $\{(1,0,0), (0,1,0), (0,0,1), (1,1,1)\}$ in R^3 form a linerly dependent set but any three of these are linearly independent. 13

3(b). Determine whether the polynomial $p(t) = t^2 + 4t - 3$ belongs to the space spaned by the polynomials $p_1(t) = t^2 - 2t + 5$, $p_2(t) = 2t^2 - 3t$ and $p_3(t) = t + 3$ or not? If yes, express $p(t)$ as a linear combination of $p_1(t)$, $p_2(t)$, and $p_3(t)$. 12

3(c). If S is a non empty subset of a vector space V over a field of K, then show that L(s) is a subspace of V containing S. 10

4(a). Find the condition on a,b,c so that the vector (a, b, c) of R^n belongs to the space generated by $u = (2,1,0)$, $v = (1,-1,2)$ and $w = (0,3,-4)$. 10

Let $\cup$ *and* W be the subspaces of R^4. 10

4(b). $\cup = L\{(1,1,0,-1), (1,2,3,0), (2,3,3,-1)\}$,
$W = L\{(1,2,2,-2), (2,3,2,-3), (1,3,4,-3)\}$.
Find the basis and dimension of (i) $\cup$ and (ii) W.

4(c). Find bases for the row space, the column space, and the null space of the matrix 15

$$A = \begin{bmatrix} -2 & -5 & 8 & 0 & -17 \\ 1 & 3 & -5 & 1 & 5 \\ 3 & 11 & -19 & 7 & 1 \\ 1 & 7 & -13 & 5 & -3 \end{bmatrix}$$

SECTION – B

5(a). Find the arc length of the curve traced out by the endpoint of the vector-valued function $\overline{r}(t) = < t^2, \cos t + t\sin t\ , \sin t - t\cos t >$, for $0 \le t \le \pi$. 09

5(b). Sketch the curve traced out by the following vector- valued functions :

i) $\vec{r}(t) = \sin t\,\hat{\imath} - 3\cos t\,\hat{\jmath} + 2t\hat{k},\ t \ge 0.$ 12

ii) $\vec{r}(t) = (1 + \cos t)\,\hat{\imath} - (3 - \sin t)\hat{\jmath},\ ;0 \le t \le 2\pi.$

5(c). A particle moves so that its position vector is given by $\vec{r}(t) = \cos\omega t\,\hat{\imath} + \sin\omega t\,\hat{\jmath}$, where $'\omega'$ is a constant. Show that 14

i) the velocity of $\vec{v}$ of the particle is perpendicular to $\vec{r}$,

ii) the acceleration $\vec{a}$ is directed towards the origin and

iii) $\vec{r} \times \vec{v} = a$ constant vector

6(a). Find the acute angle between the surfaces $z = x^2 + y^2$ and $z = \left(x - \frac{\sqrt{6}}{6}\right)^2 + \left(y - \frac{\sqrt{6}}{6}\right)^2$ at the point $p\left(\frac{\sqrt{6}}{12}, \frac{\sqrt{6}}{12}, \frac{1}{12}\right)$ 13

6(b). Interpret the physical significance of curl graphically if $\vec{v} = \vec{\omega} \times \vec{r}$, then prove that curl $\vec{v} = 2\vec{\omega}$, where $\vec{\omega}$ is a constant vector. 10

6(c). Find the directional derivative of $f = x^2 + y^2 + z^2$ at (1, 2, 3) in the direction of the line $\frac{x}{3} = \frac{y}{4} = \frac{z}{5}$. Find the maximum rate of increase of f at (1, 2, 3). 12

7(a). A particle moves along the space curve $x = 3t - t^3,\ y = 3t^2, z = 3t + t^3$. Find (i) the unit tangent vector $\hat{T}$, (ii) unit normal vector, (iii) Binorml, and (iv) Curvature. 14

7(b). A fluid motion is given by $\vec{v} = (y\sin z - \sin x)\,\hat{\imath} + (x\sin z + 2yz)\hat{\jmath} + (xy\cos z + y^2)\hat{k}$. Is the motion irrotational? If so, find the velocity potential. 12

7(c). If $F = xy\,\hat{\imath} - z\,\hat{\jmath} + x^2\,\hat{k}$ and C is the curve $x = t^2,\ y = 2t,\ z = t^3$ from $t = 1$, evaluate $\int_C \bar{F} X\, d\bar{r}$. 09

8(a). Apply Stoke's theorem to evaluate $\oint_C \vec{F}.\overrightarrow{dr}$, where $\vec{F} = y^2\hat{\imath} + x^2\hat{\jmath} - (x + z)\hat{k}$ and C is the boundary of triangle with vertices at $(0,0,0), (1,0,0)$ and $(1,1,0)$. 10

8(b). Verify Divergence theorem, given that $\vec{F} = 4xz\hat{\imath} - y^2\hat{\jmath} + yz\hat{k}$ and S is the surface of the cube bounded by planes $x = 0,\ x = 1,\ y = 0,\ y = 1,\ z = 0$ and $z = 1$. 15

8(c). Use the transformation $u = x,\ v = z - y,\ w = xy$; Evaluate $\iiint_Q (z - y)^2 xy\,dv$, where Q is the region bounded by the surface $x = 1,\ x = 3,\ z = y,\ z = y + 1,\ xy = 2$ and $xy = 4$. 10

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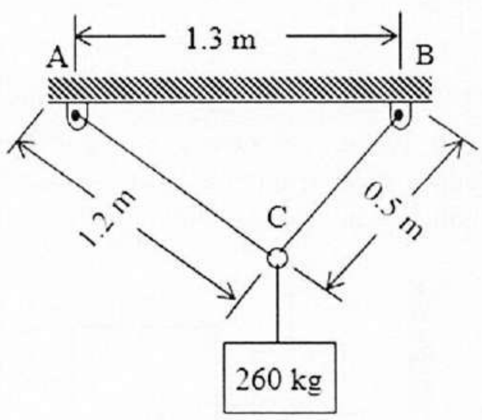
KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 2nd Year 1st Term Examination, 2022
ME 2113
(Statics and Solid Mechanics)

Time: 3 Hours. Full Marks: 210

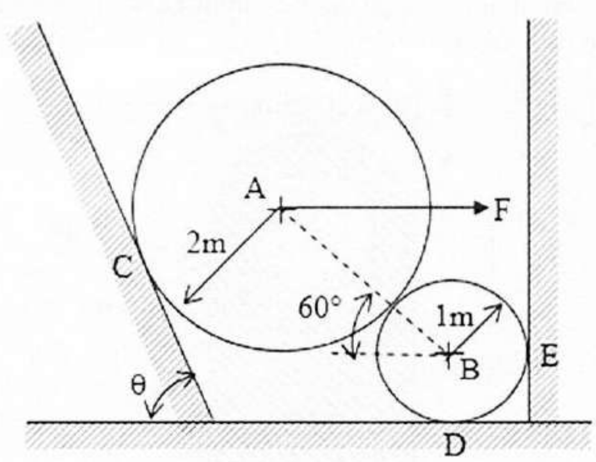
N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

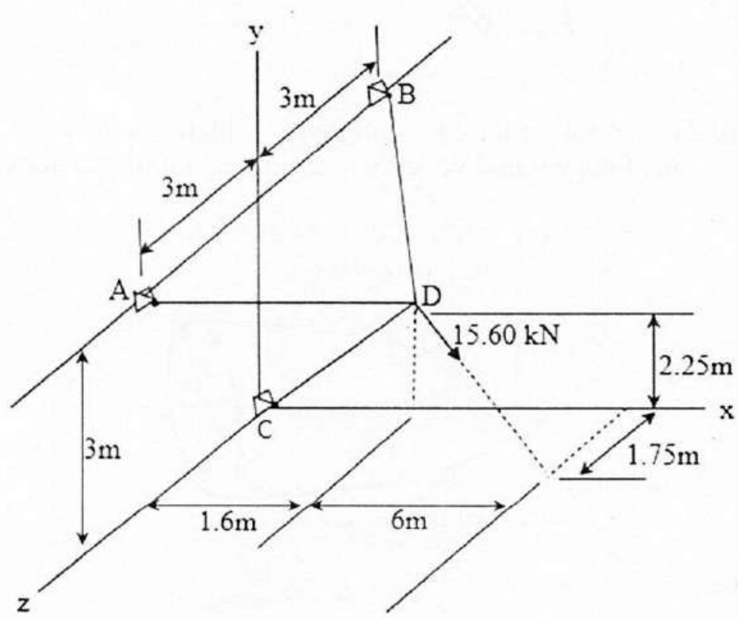
1(a). Two cables are tied together at C and loaded as shown in the figure. Determine the tension in AC and BC. 17



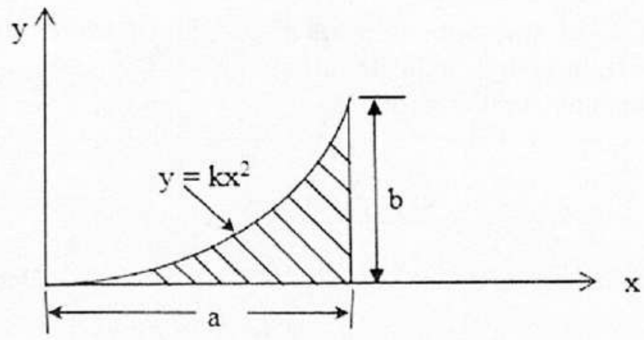
1(b). Two spheres are at rest against smooth surfaces as shown in figure. Sphere A weighs 1450 kg and sphere B weighs 180 kg. Let, the horizontal force F is 4460 N and $\theta = 75°$. Find the reactions at C, D and E. 18



2(a). Three cables are connected at D where a 15.60 kN force is applied as shown. Determine the tension in each cable. 17

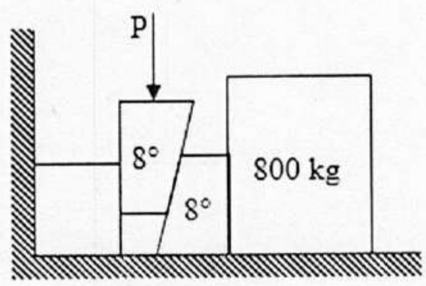


2(b). Determine by direct integration the location of the centroid of a parabolic spandrel as shown in figure. Also calculate its volume. 18

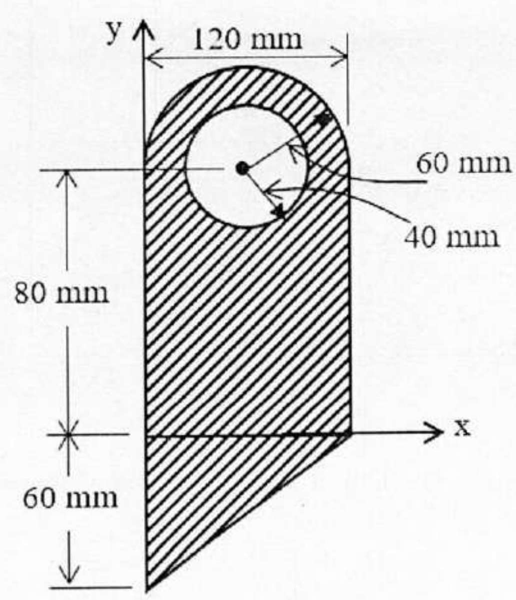


3(a). Derive an expression for ratio of tensions in two sides of a flat belt. 17

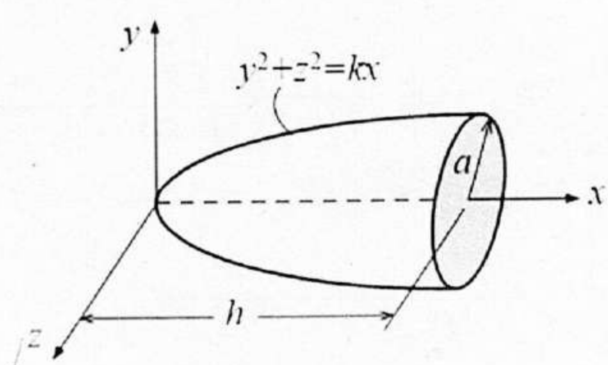
3(b). Two 8° wedges of negligible weight are used to move and position the 800 kg block. Knowing that the coefficient of static friction is 0.30 at all surfaces of contact, determine the smallest force P that should be applied as shown to one of the wedges. 18



4(a). For the plane area shown, determine (i) the first moments with respect to x and y axes, and (ii) the location of the centroid. 17



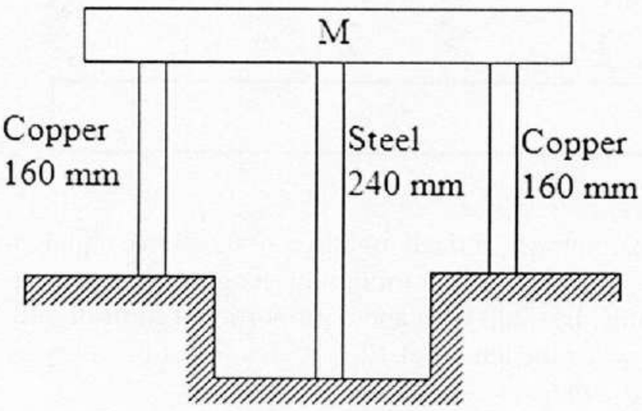
4(b). Determine by direct integration the mass moment of inertia and radius of gyration with respect to the x--axis of the paraboloid shown, assuming a uniform density and a mass m. 18



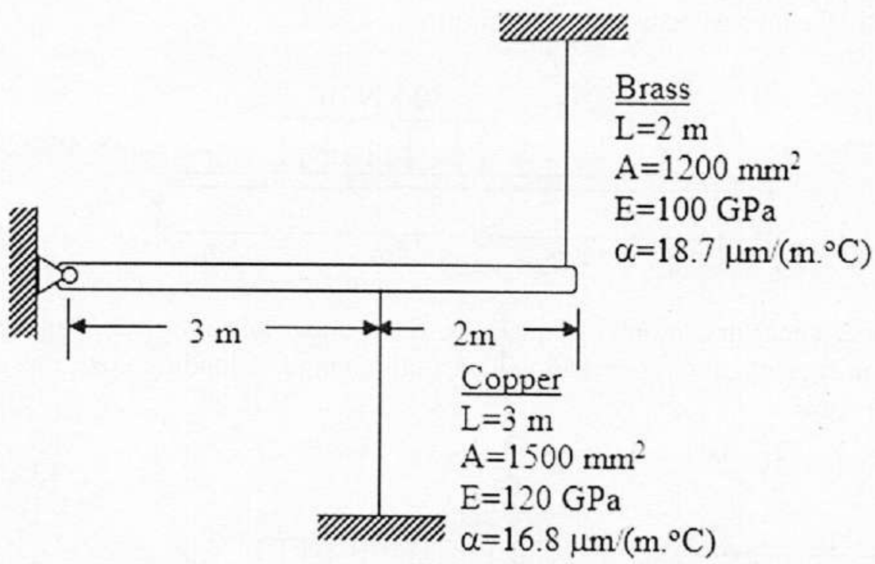
SECTION – B

5(a). Define: Engineering stress, True stress and Bearing stress. 05

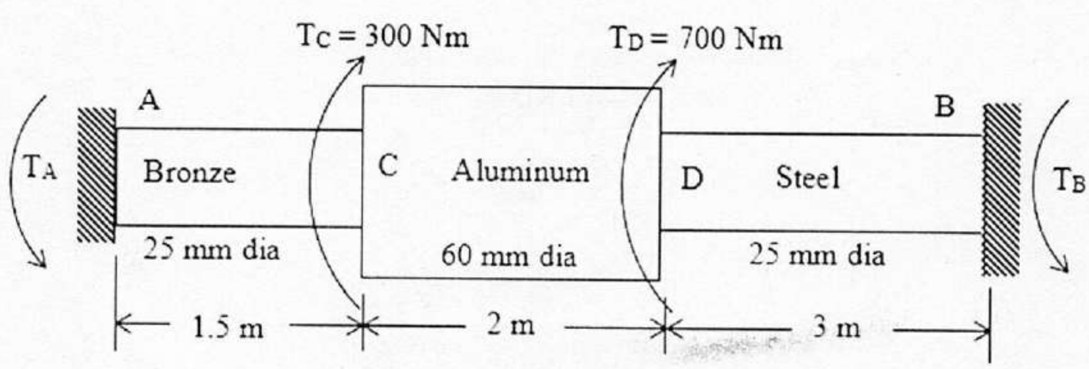
5(b). A rigid block of mass M is supported by three symmetrically spaced rods as shown in figure. Each copper rod has an area of 900 mm^2, E = 120 GPa and the allowable stress is 70 MPa. The steel rod has an area of 1200 mm^2, E = 200 GPa, and the allowable stress is 140 MPa. Determine the largest mass M, which can be supported. 15



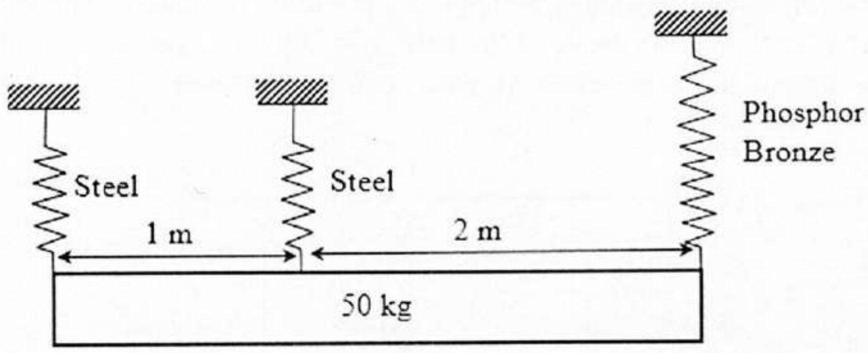
5(c). A rigid horizontal bar of negligible mass is connected to two rods as shown in figure. If the system is initially stress-free, calculate the temperature change that will causes a tensile stress of 90 MPa in the brass rod. Assume that both rods are subjected to the change in temperature. 15



6(a). A shaft compressed of segments AC, CD and DB is fastened to rigid supports and loaded as shown in figure. For Bronze, G = 35 GPa; for aluminum, G = 28 GPa and for steel, G = 83 GPa. Determine the maximum shearing stress developed in each segment. 17



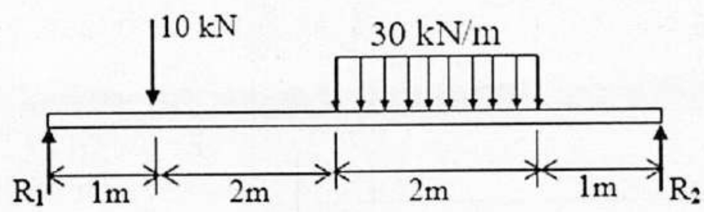
6(b). As shown in figure, a homogeneous 50 kg rigid block is suspended by the three springs whose lower ends were originally at the same level. Each steel spring has 24 turns of 10 mm diameter wire on a mean diameter of 100 mm and G = 83 GPa. The bronze spring has 48 turns of 20 mm diameter wire on a mean diameter of 150 mm and G = 42 GPa. Compute maximum shearing stress in each spring. 18



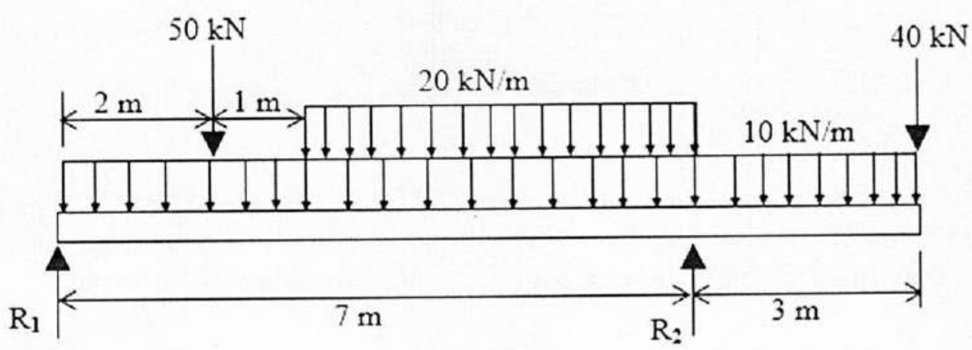
7(a). Two C 310 × 45 channels are latticed together, so they have equal moment of inertia about the principal axes. Determine the minimum length of a column having this section assuming pinned ends, E = 200 GPa and a proportional limit of 240 MPa. What safe load will the column carry for the length of 12 m with a factor of safety of 2.5. For C 310 × 45, take $I_x = 67.3 \times 10^{-6} m^4$. 18

7(b). Select the lightest W shape that can be used as a column to support an axial load of 420 kN, on an effective length of 4m. Use AISC specification with $\sigma_{yp} = 250$ MPa and E = 200 GPa. 17

8(a). Write the shear and moment equations and draw the shear force and bending moment diagrams for the beam loaded as shown in figure. 18



8(b). Without writing shear and moment equations, draw shear and moment diagram for the beam shown in figure. Give numerical values at all change of loading positions and at all points of zero shear. 17



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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 2nd Year 1st Term Examination, 2022
CSE 2113
(Computer Programming)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a). Define computer and computer programs. Briefly explain the types of translator program and also explain why every programming language need a translator program. 08

1(b). What is an IDE and what are the basic features commonly found in IDES that facilitate code editing and development? Explain the purposes of different MATLAB windows. 09

1(c). Explain the relation between data and variable name with an example. What is meant by MATLAB key words? Write the basic rules for naming variables in MATLAB with example. 06

1(d). Write a short note on the following symbols used in MATLAB- 12

i) Semicolon(;), ii) percent(%), iii) clc, clear, iv) fprintf, v) input, iv) format

2(a). Define data and datafile with examples. Mention different data types used in MATLAB language and how do you create a character array in MATLAB. 10

2(b). Explain the steps those are needed to write the output in a file. 10

2(c). Define variables, x=0.85, y=12.5, and then use them to create a column vector that has following elements: 06

$\frac{\sin 65°}{\sin 80°}$, y, y^x, $\ln\left(\frac{x}{y}\right)$, x$\times y$, and $\sqrt{x+y}$

2(d). The spread of a computer virus through a computer network can be modeled by $N(t) = 20e^{0.15t}$ where N(t) is the number of computers infected t time in munutes. Write a MATLAB script to 09

i). Determine how long it takes for the number of infected computers to double.

ii). Determine how long it takes for 1,000,000 computers to be infected.

3(a). Define the system of linear equations. Explain with an example of the representation of a linear system into vector equation of the system and Matrix equation of the system. Derive a system of equations to find the partial fraction decomposition of the given rational expression. Then write MATLAB code to solve the system 12

$$\frac{4x^2}{(x+1)^2(x-1)} = \frac{A}{(x-1)} + \frac{B}{(x+1)} + \frac{C}{(x+1)^2}$$

i) By considering the augment Matrix of the system.
ii) By considering the coefficient Matrix of the system.

3(b). Define SVD and PLU decomposition with mathematical relation and write an example code to find them using MATLAB with the reconstruction of the original Matrix. Also, Discuss the different types of SVD. 10

3(c). Write a MATLAB code for creating a 4×4 random matrix. Use built in function to find eigen vectors and eigen values of that matrix. Also check if the eigen vectors are linearly independent and span $\mathbb{R}^4$. 13

4(a). What is meant by function in MATLAB? Briefly explain how to create and call anonymous and user-defined functions in MATLAB with examples. 11

4(b). Define local and global variables with MATLAB examples. Write an anonymous MATLAB function for the following math function: 12

$y(x) = 0.6x^3e^{-0.47x} + 1.5x^2e^{-0.6x}$

The input to the function is x and the output is y. Write the function such that x can be a vector (use element-by-element operations)
i) Use the function to calculate y(-2) and y(4).
ii) Use the function to calculate y(x) for $-4 \le x \le 8$.

4(c). Write an function , ' y= is prime(x)' to check whether x is prime (y=1 if x is prime) or not (y=0 if x is not prime). Using this function write a program to find first 50 prime numbers, store these prime numbers in variable-prime numbers, and display these numbers. 12

SECTION – B

5(a). With proper example and syntax explain the function of the following command: 09

i). Magic, ii). find , iii). Cumsum

5(b). With proper example explain how meshgrid command transforms two different vectors into array of the same size? 15

5(c). A matrix is given as; 08

$$M = \begin{bmatrix} 3 & 8 & 9 \\ -5 & 0 & 11 \\ 5 & 2 & 4 \end{bmatrix}$$

What will be the output of the following command?
i). $a = M(:, length(M): -1: 1)$
$ii).\ a_1 = M(find\ (M > 4))$
$iii).\ b = fliplr(flipud(M))$
$iv).\ c = cumsum(sum(M))$

5(d). Writing a single line command create the following matrix: 03
[Use eye, ones and zeros commands to create the matrix]

$$D = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

6(a). Why do line command use in 2D plotting? With proper syntax explain how line command is used? 10

6(b). With proper syntax explain the function of the following plotting command: 09
$i).\ Semilogx$, ii) errorbar, iii) fplot

6(c). Two parametric equations are given below by : 12

$x = cos^3(t),\ \ y = sin^3(t)$
$u = \sin(t),\ \ \ v = \cos(t)$

In one figure, make a plots of y versus x and v versus u for $0 \le t \le 2\pi$. Format the plot such that the both axes will range from $-2\ to\ 2$.

6(d). What is line specifiers? What are the things included in line specifiers. 04

7(a). With proper syntax explain the function of polyder command. 09

7(b). An experimental data fits the following function: $y = bx^m$ 05

Briefly explain how will you determine the value b and m.

7(c). The relationship between two variables p and t is known to be : 15

$$p = \frac{mt}{b + t}$$

The following data points are given:

t	1	3	4	7	8	10
p	2.1	4.6	5.4	6.1	6.4	6.6

Determine the constant m and b by curve fitting the equation.

7(d). Make a 3-D surface plot of the function $z = \frac{y^2}{4} - 2\sin(1.5x)$ in the domain $-3 \le x \le 3$ and $-3 \le y \le 3$. 06

8(a). Suppose a code is written as : 10

$x = [20, 25];$

$y = [100,120];$

$y_1 = 112;$

$x_1 = interp1(x, y, y_1, 'nearest')$

$x_2 = interp1(x, y, y_2\ , Linear')$

$x_3 = interp1(x, y, y_1, 'Spline')$

$x_4 = interp1(x, y, y_1, 'Pchip')$

Is there any error in the code? If yes then by rectifying the error, write down the correct code and explain the output of the code.

8(b). Draw the flow chart for if elseif-else-end structure and describe program for this construct with MATLAB code. 08

8(c). Rectify the error in the following code and write down the proper code. 06

i) $s = input("Enter\ a\ text\ string:");$

$if\ (s == "Hi")$

$disp\ ("\ you\ entered\ Hi");$

$else\ if$

Disp(" you did not enter Hi")

end

ii) $m = 1$

$While\ 1$

$Disp('CSE\ 2113')$

end

8(d). The reciprocal Fibonacci constant Ψ is defined by the infinite sum: $\Psi \sum_{n=1}^{\alpha} \frac{1}{F_n}$ 11

Where F_n are the Fibonacci numbers 1, 1, 2, 3, 5, 8, 13, ... Each element in this sequence of numbers is the sum of the previous two. Start by setting first two elements equal to 1, then $F_n = F_{n-1} + F_{n-2}$. Write a program in a script file that calculates Ψ for a given n. Excute the program for n= 10 and 100.

......End......